MILITARY SPECIFICATION

FLASHLIGHTS, PENLIGHT STYLE, PILOTS

This specification has been approved by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers small dry-cell operated flashlights which provide both red and white light.

1.2 Classification. The flashlights shall be furnished in the following part numbers, as specified (see 6.2):

MS21998-1 - without lanyard
MS21998-2 - with lanyard

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

	W-L-00111/11	Lamp, Incandescent, Miniature Trade No. 222, Focused, Miniature Screw Base.
	PPP-B-566	Box, Folding, Paperboard.
*	PPP-B-601	Boxes, Wood, Cleated Plywood.
	PPP-B-636	Boxes, Shipping Fiberboard.
*	PPP-B-676	Boxes, Setup.

MILITARY

*	MIL-B-18/7	Battery, Dry, BA-27.
	MIL-P-116	Preservation, Method of.
	MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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*	DOD-D-1000	Drawing, Engineering and Associated List.
*	MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
	MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

FEDERAL

*	FED-STD-3	Color, Aeronautical Lighting.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods.
MS21998	Flashlight.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adoption shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 633	Zinc on Iron and Steel, Electrodeposited Coatings of.
ASTM D 3951	Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. This specification makes provisions for first article inspection (see 4.3.1 and 6.2).

3.2 Components. The flashlight components shall consist of a lamp head, battery case or housing, pocket clip, filter, lamp, and switch in accordance with MS21998. Unless otherwise specified, batteries shall not be furnished with the flashlight (see 6.2).

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3.3 Materials.

3.3.1 Protective treatment. Where materials are used in the construction of the flashlight that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.2 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.2.1 Standard parts. Standard parts (MS) shall be used whenever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the standard parts (MS) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids or request for proposal, commercial parts may be used, provided they conform to all requirements of this specification.

* 3.3.2.2 Reclaimed materials. Reclaimed materials shall be used to the maximum extent possible without jeopardizing the intended use of the item.

3.4 Design and construction. The design and construction shall be in accordance with MS21998.

3.4.1 Construction. The flashlight shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1.1 The flashlight shall be so constructed that lamp and battery replacement can be easily made by personnel of operating units and overhaul bases without the use of tools.

3.4.2 Lamp head. The lamp head shall include the lamp socket, light shield, and adjustable filter. Where the lamp head is made of nonmetallic material, a metal insert or sleeve with screw threads shall be provided for the junction with the case.

3.4.2.1 The head shall be designed to accommodate a miniature screw-base lamp. It shall be so designed that the lamp may be easily replaced.

3.4.2.2 The head shall be provided with a suitable opaque shield or housing that will cover the lamp. The end of the opaque shield or housing shall be provided with a 0.2 inch minimum hole in the outer end. The hole shall coincide with and shall be approximately the same diameter as the lens in the end of the lamp.

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* 3.4.2.3 Filter. Each flashlight shall include an integral filter which can be moved into place over the lens of the lamp and moved away from the lens of the lamp without the use of tools and without any disassembly of the flashlight. When the filter is in place over the lens of the lamp, it shall permit at least 6 percent of the light emitted by the lamp at each point within 10 degrees of the axis of the lamp to leave the flashlight as aviation red conforming to category I of FED-STD-3 and shall prevent any light not conforming to aviation red of FED-STD-3 from leaving the flashlight. When the filter is away from the lens of the lamp, all of the light emitted by the lamp within 10 degrees of the axis of the lamp shall leave the flashlight in the same pattern in which it is emitted by the lamp. A detent or other approved means shall be provided to assure positive positioning of the filter.

3.4.3 Housing. The battery housing shall be fabricated from a cellulose base plastic material, injection-mold type, or equal, or a lightweight corrosion-resistant metal. Where the case is made of nonmetallic material, a metal insert or sleeve with screw threads shall be provided for the junction with the lamp head.

* 3.4.3.1 The housing shall be designed to house two size AA flashlight batteries in accordance with MIL-B-18/7.

3.4.4 Switch and pocket clip. The flashlight shall be provided with a suitable "OFF-ON" switch and suitable pocket clip. The design may be such that the switch and pocket clip are an integral unit or may be separate.

3.4.4.1 Switch. The switch shall be of such design that the flashlight may be held and turned to the "ON", or "OFF" position with one hand and cannot be accidentally energized when clipped in the pocket and carried normally.

3.4.4.2 Switch positions. The switch positions "OFF" and "ON" shall be as shown on MS21998. A detent or other approved means shall be provided to assure positive positioning of the switch in the "ON" and "OFF" positions and prevent accidental or inadvertent dislocation.

3.4.4.3 Pocket clip. The clip shall maintain adequate pressure against the housing to assure retention of the flashlight in pockets made of lightweight material when the pockets are inverted. Clip location and dimensions shall be as shown on MS21998.

3.4.5 Lamp. Each flashlight shall be furnished with one lamp in accordance with W-L-00111/11.

3.4.6 Lanyard. For the flashlight with lanyard, a suitable eye at the rear end of the flashlight and an adjustable lanyard shall be provided. The lanyard shall be adjustable so that the cord making the maximum loop shall measure approximately 41 inches when subjected to a pull of at least 2 pounds.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.6 Finishes and protective coatings.

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3.6.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows.

3.6.1.1 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.6.1.2 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.6.1.3 Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

* 3.6.2 Zinc plating. Zinc plating shall be in accordance with ASTM B 633-78.

3.7 Weight. The weight of the flashlight without batteries shall not exceed that shown on MS21998.

3.8 Dimensions. The flashlight shall not exceed the dimensions shown on MS21998.

3.9 Performance. The flashlight shall be capable of withstanding the following conditions without failure:

a. Operation: As specified in 4.6.2.

b. Temperature: A temperature range of -30° to 110° F, as specified in 4.6.3 and 4.6.4.

c. Humidity: A relative humidity of 95 $\pm$5 percent at a temperature of 71° $\pm 2^{\circ}$ C, as specified in 4.6.5.

d. Fungi: Moist fungi encountered in tropical and semitropical climates, as specified in 4.6.6.

e. Salt spray: As specified in 4.6.7.

f. Impact: As specified in 4.6.8.

g. Lanyard pull: As specified in 4.6.9.

h. Filter color: As specified in 4.6.10.

i. Pocket clip: As specified in 4.6.11.

j. Switch-life test: As specified in 4.6.12.

3.10 Identification of product. The following information shall be permanently and legibly marked on the assembly in accordance with MIL-STD-130:

FLASHLIGHT, PENLIGHT STYLE, PILOTS
MS part No.
Manufacturer's part No.
Manufacturer's name or trademark
U.S.

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3.11 Workmanship. The flashlight, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. It shall be free from defects, burrs, and sharp edges. Marking of parts and assemblies shall be legible. Soldering, welding, brazing, painting, and wiring shall be thorough. Parts shall be properly aligned, and assembly screws and bolts shall be tight.

3.11.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.2 Cleaning. The flashlight shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The examination and testing of the flashlight shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.6.

4.3.1 Sampling instructions. The first article samples shall consist of five flashlights representative of the first article item. Samples shall be tested in a place and manner designated in the procurement document (see 6.2).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following examination and test, and any other test specified herein which is deemed necessary to determine compliance with this specification:

a. Examination of product (see 4.6.1)

b. Operation (see 4.6.2).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with the table entitled "Sample size code letters for ordinary inspection" of MIL-STD-105, using inspection level II and an acceptable quality level (AQL) of 0.65 percent defective.

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4.4.1.1 Lot. A lot for quality conformance inspection shall consist of all flashlights made under the same conditions at the same time.

* 4.5 Test conditions. Test conditions shall be the standard conditions specified in MIL-STD-810.

4.6 Inspection methods.

4.6.1 Examination of product. The complete flashlight shall be inspected to determine compliance with the requirements specified herein with respect to materials, weight, overall dimensions, workmanship, and marking.

4.6.2 Operation. The flashlight shall be tested for proper operation.

* 4.6.3 Low temperature. The complete flashlight without batteries but with lamp, shall be subjected to low-temperature test method 502, MIL-STD-810. At the end of the test, the flashlight shall be examined. There shall be no crazing, dimensional changes, or other defects which interfere with its proper operation.

* 4.6.4 High temperature. The complete flashlight without batteries but with lamp, shall be subjected to high-temperature test method 501, procedure II, MIL-STD-810. At the end of the test, the flashlight shall be examined. The flashlight shall be compared with an acceptable untested sample for dimensional stability, crazing, and other defects. The flashlight shall be in full working condition after the test. All threads shall be workable and all parts of the light interchangeable with the untested sample.

* 4.6.5 Humidity. The flashlight, complete with batteries and lamp, shall be subjected to humidity test method 507, procedure II, MIL-STD-810. At the end of the test, the flashlight shall be examined. There shall be no crazing, dimensional changes, or other defects which interfere with its proper operation.

* 4.6.6 Fungus. The flashlight, complete with lamp but not batteries, shall be subjected to fungus-resistance test method 508, MIL-STD-810. At the end of this test, the flashlight shall be examined. There shall be no dimensional changes or defects which interfere with its proper operation.

* 4.6.7 Salt spray. The flashlight shall be subjected to salt spray test method 509, MIL-STD-810.

NOTE: On tests of 4.6.3 through 4.6.7, failure of the batteries shall not constitute a failure of the flashlight.

* 4.6.8 Impact. The flashlight, complete with batteries and lamp, shall be dropped in free fall onto a wooden floor from a height of 72 inches. The flashlight shall be dropped once on the side of the case and once on the end opposite the lamp. At the end of this test, the flashlight shall be examined. There shall be no failures of the flashlight, except for the lamp, which would interfere with its proper operation.

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4.6.9 Lanyard. The lanyard shall withstand a pull of 15 pounds when suspended from a 6 inch diameter cylinder for 30 minutes. Any separation or signs of damage shall be cause for rejection.

4.6.10 Filter color. The filter shall be tested for conformance to 3.4.2.3.

4.6.11 Pocket clip. The pressure point of the clip shall be pulled away from the housing a distance of 0.375 inch a total of 50 times. No distortion, breakage, or reduction of holding pressure against the housing shall result. A pull of 2 ounces ±1 ounce shall be required to remove a sheet of paper held between the clip and housing after the above test.

4.6.12 Switch-life test. The on-off switch shall be operated "ON-OFF-ON" 100 times. Upon completion, the switch must operate satisfactorily and show no sign of deterioration.

4.7 Packaging, packing, and marking. Preparation for delivery shall be inspected for conformance to section 5.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, or commercial, as specified (see 6.2).

5.1.1 Level A. Each flashlight shall be packaged one each, method IC-1 of MIL-P-116. Batteries, when furnished with the flashlight, shall be individually wrapped in barrier material conforming to MIL-B-121. The flashlight (and batteries when furnished) shall be unit packaged in a container conforming to PPP-B-566 or PPP-B-676. Unit packages shall be intermediately packaged in containers conforming to PPP-B-566 or PPP-B-676. The gross weight of the intermediate package shall not exceed approximately 10 pounds.

* 5.1.2 Commercial. Flashlights shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

* 5.2.1 Level A. A quantity of flashlights preserved as specified in 5.1, shall be packed in a wood, cleated plywood shipping container conforming to PPP-B-601, overseas type, style A, grade A. Insofar as practical, containers shall be uniform in shape and size, contain identical quantities, and be of minimum cube and tare consistent with the protection required. Container closure and strapping shall be in accordance with the appendix to PPP-B-601.

* 5.2.2 Level B. A quantity of flashlights preserved as specified in 5.1 shall be packed in a wood, cleated-plywood shipping container conforming to PPP-B-601, domestic type, style A, Grade B, or a fiberboard shipping container conforming to PPP-B-636, class weather resistant type optional. Insofar as practical, containers shall be uniform in shape and size, contain identical quantities, and be of minimum cube and tare consistent with the protection required. Container closure and strapping shall be in accordance with the appendix to the applicable box specification.

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* 5.2.3 Commercial. Flashlights preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate containers and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

6. NOTES

6.1 Intended use. The flashlight covered by this specification is intended for use by aircraft personnel for temporary lighting purposes.

* 6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. MS part number, as applicable (see 1.2).

c. Whether batteries should be furnished with flashlight (see 3.2).

d. Where the first article inspection samples should be sent and instructions concerning the submittal of the test reports (see 4.3.1).

e. Levels of preservation packaging and packing required (see 5.1 and 5.2).

f. First article.

6.3 Type A-6B flashlights formerly procured under this specification are now covered by the applicable MS part numbers (see 1.2).

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Air Force - 99

Reviewer activities:

DLA - GS

Preparing activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-7326G	FLASHLIGHTS, PENLIGHT STYLE, PILOTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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